Homework

1. Prove or disprove the following.

   a. A constant function is not one-to-one if the domain has more than one element.
   b. The identity function is one-to-one.
   c. The identity function is onto.

2. Let $g : A \to B$ and $f : B \to C$. Prove that if $f \circ g : A \to C$ is one-to-one and g is onto then f is also one-to-one.

3. Suppose that $f : A \to B$ is a one-to-one function. Prove or disprove that if $f^{-1} : f(B) \to A$ is a function then f^{-1} is also one-to-one.

4. Let $f : A \to B$ be a one-to-one function. Show that $|A| \leq |B|$.

5. Let $f : A \to B$ and $g : B \to C$ be functions. Prove that $g \circ f : A \to C$ is a function.

6. (This problem was dropped) Let $f : A \to A$, $g : A \to A$, and $h : A \to A$; prove that $(h \circ g) \circ f = h \circ (g \circ f)$.

7. Prove that if $f : A \to B$ is onto and $g : B \to C$ is onto, then $(g \circ f) : A \to C$ is onto.

8. Each of the following statements, $P(x, y)$, define a relation R in the natural numbers, $\mathbb{N}$, i.e., $R = (\mathbb{N}, \mathbb{N}, P(x, y))$. Show whether each of the following relations is reflexive.

   a. $P(x, y) =$"x is less than or equal to y"

   b. $P(x, y) =$"x divides y"
   c. $P(x, y) =$"$x + y = 10$"
   d. $P(x, y) =$ "x and y are relatively prime."

Presentation problems.

1. Let the function $f : A \to B$ be 1-1 and onto, i.e., f^{-1} exists and $|A| = |B|$. Prove that $\left(f^{-1} \circ f\right) : A \to A$ is the identity function on A and that $\left(f \circ f^{-1}\right) : B \to B$ is the identity function on B.

2. Let $f : A \to B$ be 1-1 and onto. Prove that $g = f^{-1}$, if $(g \circ f) : A \to A$ is the identity function on A and $(f \circ g) : B \to B$ is the identity function on B.

3. The *product set* of sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$; it is denoted by $A \times B$. Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

4. Prove: Let R and R' be symmetric relations in a set A; then $R \cap R'$ is a symmetric relation in A.

5. Prove that $\displaystyle\sum_{k=1}^{n} 2k = n^2 + n$

6. The Fibonacci numbers are :$1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$. In general, the Fibonacci numbers are defined by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_{n+1} + f_n$ for $n = 1, 2, 3, \ldots$. Prove that the n^{th} Fibonacci number $f_n < 2^n$.

7. Prove that there infinitely many prime numbers.

☺. Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than B if there is a one-to-one function from A into B, but there does not exist a bijective function from A into B; we write$|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.